DATANOVA 2026 · PROBLEM STATEMENT DN-BU-01

Lighthouse

Churn prediction and causal retention planning

See who is about to leave, what each customer is worth, and the offer that will actually keep them.



Team Tiramisu Tensor

Live app lighthouse-retention.vercel.app **Code** github.com/ac265640/LightHouse

The problem we chose

DN-BU-01

Customer Churn Prediction and Retention Strategy

The brief asks for three things

- A churn prediction model
- Insights about customer segments
- A retention playbook the business can act on

OUR DATA

7,043

real customers of a California telecom, from the IBM Telco Customer Churn 11.1.3 dataset

- Services
- Billing
- Offers received
- Exit reasons
- Location
- 52 columns

One quarter of history, so time has to be modelled carefully.

1 in 4

customers walked out in a single quarter

1,869 customers lost, out of 7,043

$139k of monthly revenue gone with them

$389k of lifetime value still at risk in those who stayed



Every dot below is a real customer at their real location. Red dots left last quarter.

Why most churn projects never pay off

A high accuracy score is where the hard questions start. Three traps sit between a good model and a retention program that makes money.

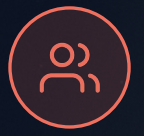


Models that cheat

Some columns are only written after a customer leaves. Keep them and the model looks perfect in testing, then fails in real use.

0.998

accuracy of a model that cheats



Risky isn't persuadable

The riskiest customers often leave whatever you offer, and many others would have stayed anyway. Ranking by risk wastes money.

−$11.5k

the usual approach, on a $15k budget



No sense of time or value

A yes-or-no score can't say how many months a customer has left, so it can't price a save or say who to call first.

1 quarter

of data: time must be modelled

Lighthouse answers three questions



Who is leaving, and why?

Five model families compete, then a calibrated ensemble scores every customer, with the top five reasons for each.



When, and what are they worth?

A survival model gives the months each customer has left, and turns them into lifetime value and value at risk.



What will actually keep them?

Causal inference measures how much each offer really changes churn, and a budget optimiser chooses who to contact.

The output

A ranked call list

Who to contact, with which offer, and the expected return.



Delivered as a live product, not a notebook

A web workspace for retention analysts, an API that scores new customers, and a Gemini copilot that answers from the real numbers.

Step 1 · We refused to let the model cheat



These six columns are only written after a customer leaves. We tested every column alone and removed all six.

~~0.998~~

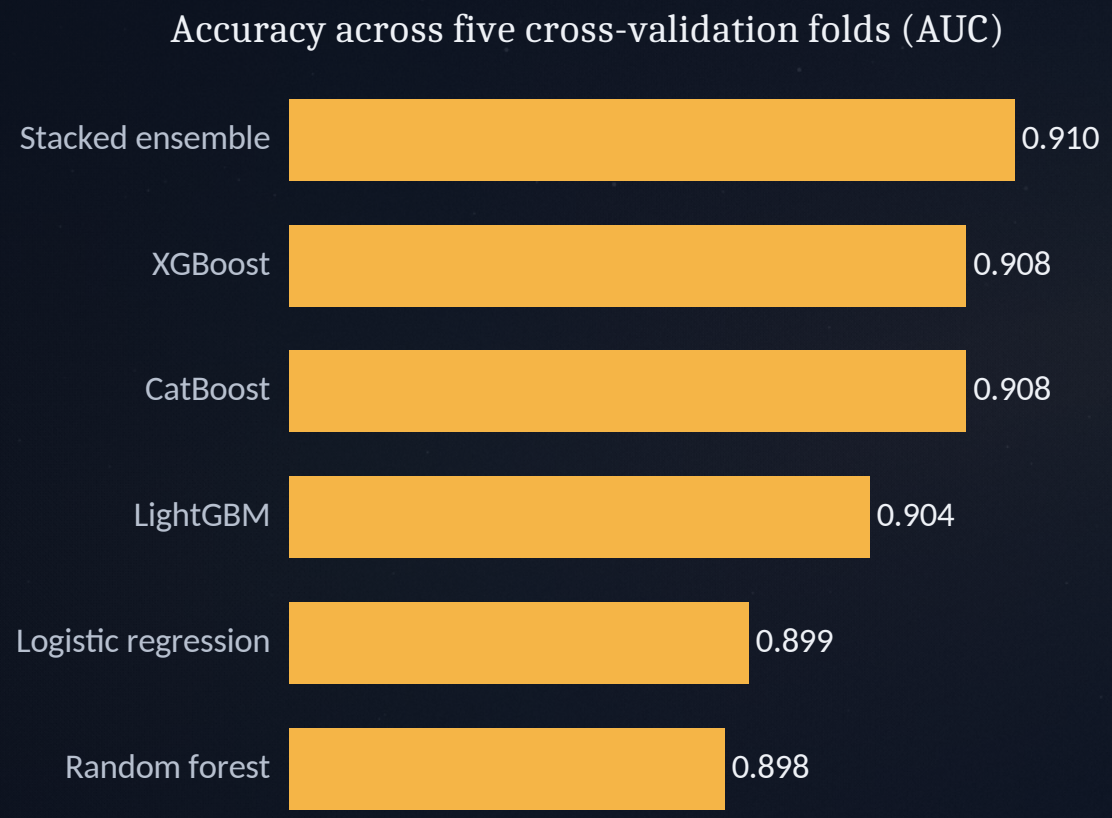
with the leaks: looks perfect, fails in real use

0.923

honest score on 1,409 customers it never saw

Gender is kept out of the model and used only to audit fairness.

Step 2 · Five models compete, the best is calibrated



The ensemble combines all five, then is calibrated so its probabilities are honest.

0.923
accuracy (AUC) on unseen customers

1.4%
calibration error: 30% really means 30 in 100

3.4×
more churners in the top 10% than average

93%
coverage for its confidence labels

Step 3 · How long each customer has, and their worth

The data is a one-quarter snapshot, so everyone in it has already survived to today. A left-truncated survival model corrects for that, giving every customer a timeline and a dollar value.

BEFORE THE FIX

87% risk this quarter, yet 52 months left. A contradiction.

AFTER THE FIX

Two separate models agree on who is at risk (correlation 0.95).



0.83
survival accuracy (c-index)

0.90
accuracy on this quarter's churn

Example: customer 5285-MVEHD

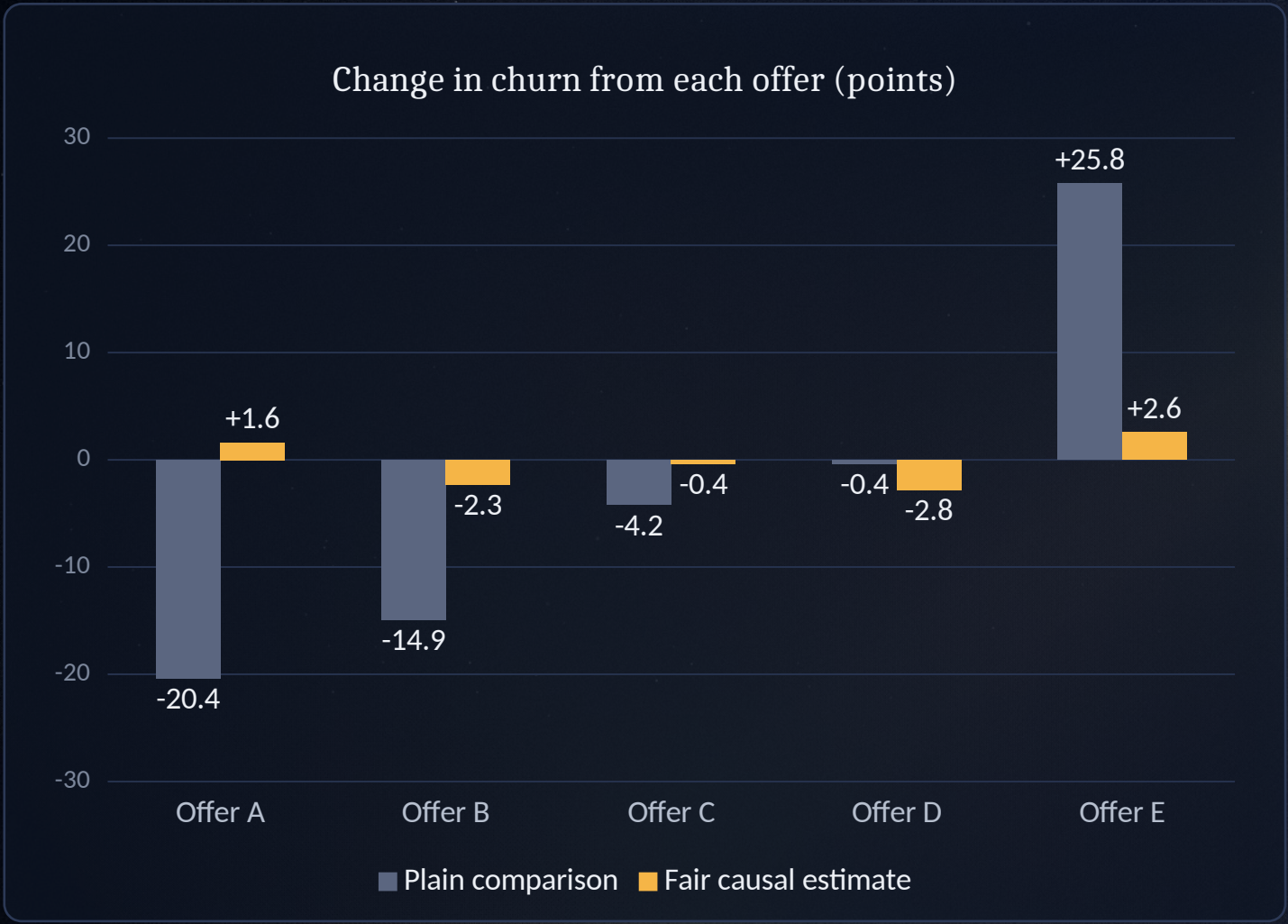
59% churn risk

14 months left

$1,150 worth if kept

$678 value at risk

Step 4 · The offer that looked best did nothing



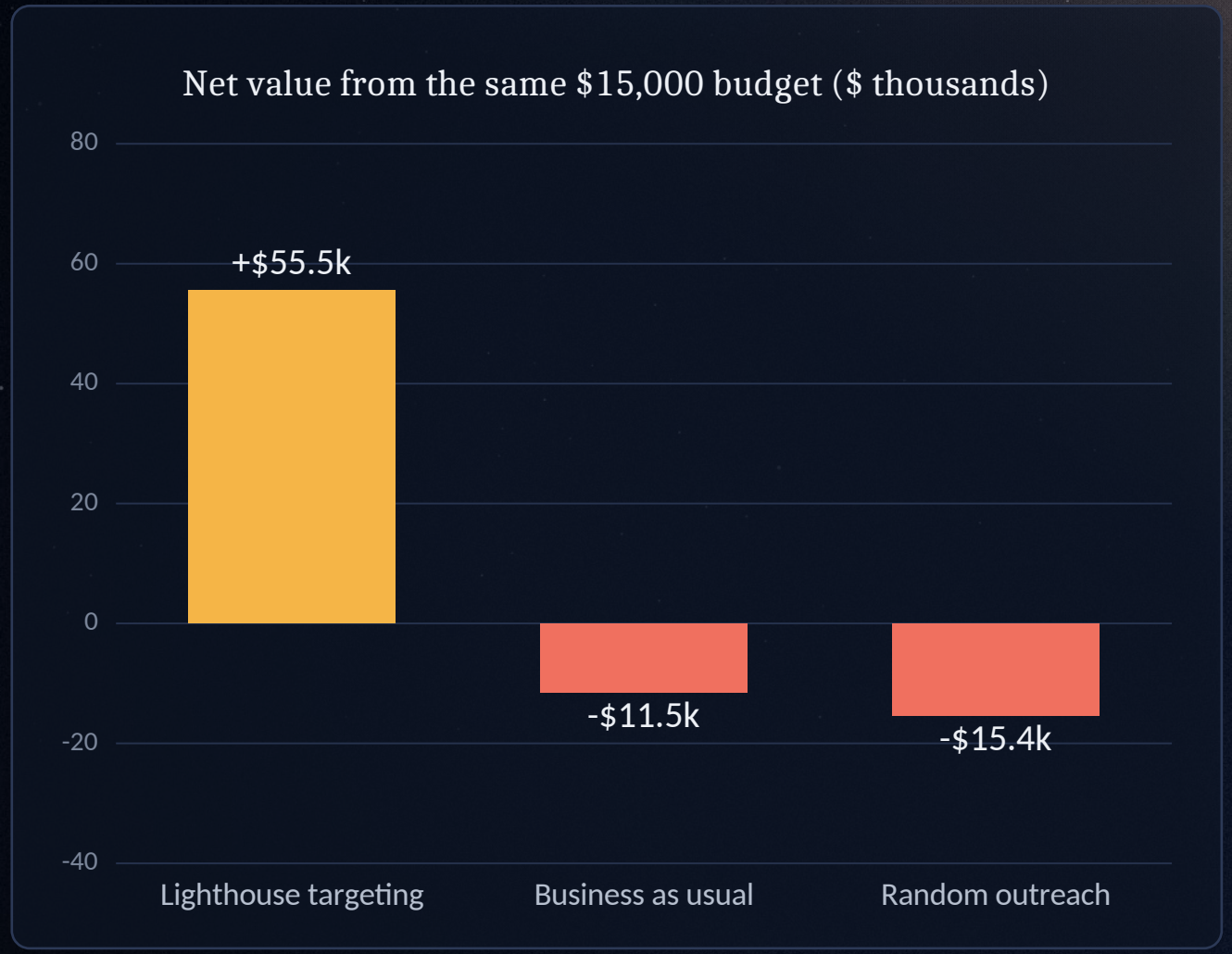
The twist

A plain comparison says Offer A cuts churn by 20 points. But Offer A only reaches customers of 66+ months, who rarely leave anyway.

WHAT WE FOUND

- Compared fairly, Offer A's effect is +1.6 points.
- No offer works on average. The value is in who receives it.
- Every campaign ships with an A/B test size.

Step 5 · Spend where it changes the outcome



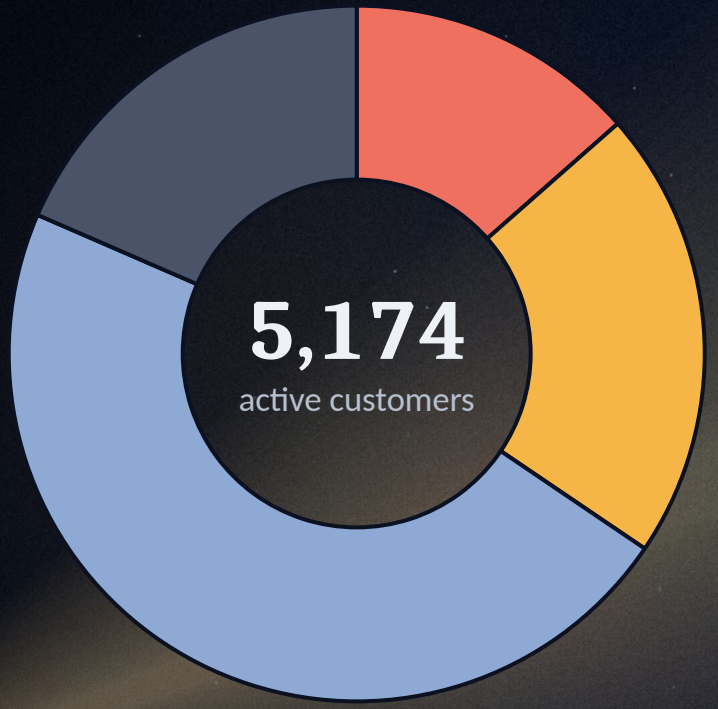
3.7×

return on the campaign, where the usual approach loses money

How the optimiser decides

- Score each customer and offer: churn prevented × lifetime value − cost
- Rank everyone, spend the budget from the top
- 400 customers contacted, about 65 saved

Segments and the retention playbook



- **Save now** 699 call within 7 days
- **Low-cost save** 1,085 automated digital touch
- **Grow** 2,434 upsell and referrals
- **Maintain** 956 standard communications

Each at-risk customer gets a play from their own drivers

Contract lock-in

Discounted 12 or 24-month contract, priced from their value

1,227 customers

Community & referral

A free month per friend referred, family plan to add lines

444 customers

Early-life onboarding

Welcome call and setup help in the first 90 days

55 customers

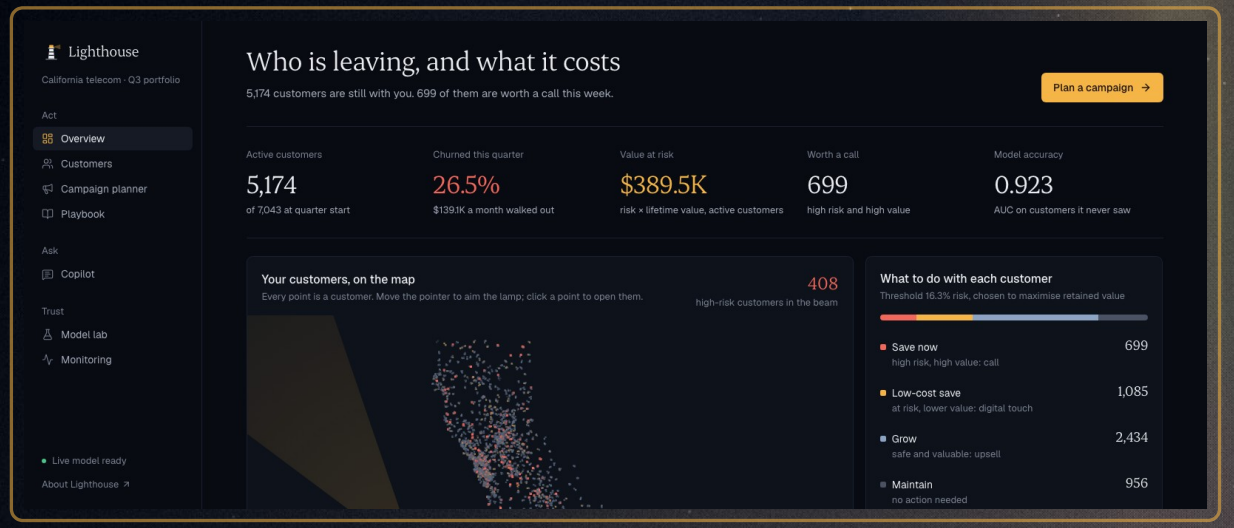
Value reframe

Right-size the plan when the bill is the problem

38 customers

Seven plays in total, each with actions, a channel and a KPI to track.

A live workspace for retention teams





Live customer map

Every customer at their real location, lit by risk

Profile and what-if

Change a contract and the risk moves: 59% → 29%

AI retention email

Gemini writes from each customer's own drivers

Campaign planner

Budget split, three strategies, A/B size, CRM export

AI copilot

Answers from Lighthouse's data and shows its sources

Model lab and monitoring

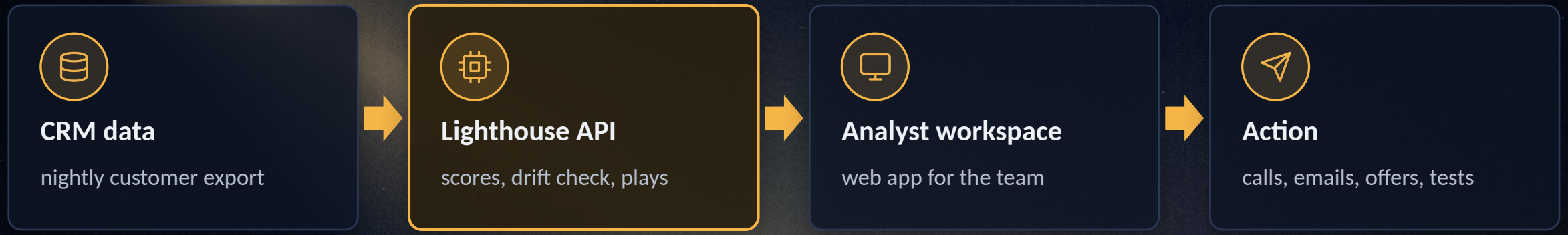
Evidence, fairness audit and drift checks

Try it live: lighthouse-retention.vercel.app

How Lighthouse is different

	A typical churn project	Lighthouse
Testing	Random split, leaky columns left in	Leakage audit and a locked holdout of unseen customers
Probabilities	Scores that only rank customers	Calibrated: 30% means 30 in 100, plus confidence labels
Time and value	Will they leave: yes or no	Months left and lifetime value for every customer
Offers	Assume the offer works	A causal estimate of what each offer really changes
Decision	A list of the riskiest customers	A budget plan with expected return and an A/B test size
AI	A chatbot that can guess numbers	A copilot that only uses the model's own numbers
Delivery	A notebook	A live web app and API, 30 tests, $0 hosting

Built to run in a real company



ROLLOUT PLAN

Weeks 1–2

Score the nightly export

Connect the CRM file, check drift first, send call lists to the team

Months 1–2

Prove the offers

Run the A/B tests the planner sizes, retrain on the results

Quarter 2

Close the loop

Trigger plays automatically when risk crosses the threshold

Works for any subscription business: telecom, streaming, SaaS, insurance and banking. It runs today on free hosting.

Built to be trusted, honest about its limits

Fairness audit

Recall is checked by gender, age and seniority. Gaps are shown in the app, and gender is never a model input.

Drift monitoring

Every feature is compared with training data. New files get a stable, watch or drift verdict before anyone acts.

Tested and reproducible

30 automated tests, one command to retrain everything, and a read-only AI with usage limits.

What we are honest about

- One quarter from one company: retrain on a client's own history first.
- Offer effects are observational, so every campaign is A/B tested.
- Exit reasons can't be predicted from the profile (44% vs a 45% baseline), so plays use churn drivers.
- Recall differs by group, and the fairness audit shows it.



Lighthouse turns a churn score into a plan

Who to save, what they're worth, and the offer that will actually keep them.

0.923	1.4%	0.83	3.7×
honest accuracy	calibration error	survival accuracy	campaign return

Live app lighthouse-retention.vercel.app
Videos lighthouse-retention.vercel.app/watch
Code github.com/ac265640/LightHouse

Thank you

Team Tiramisu Tensor